

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2021-2022
FULL MARKS: 150

CSE 4307: Introduction to Database Management System

Programmable calculators are not allowed. Do not write anything on the question paper. Answer all 6 (six) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. a) Differentiate between relation and relation instance. "A relation may not have any superkey" - briefly mention when this situation may occur and state the associated problem. 5
(CO1)
(PO1)
- b) Suppose you are going to automate a typical private company's human resource management system. How will you select the format of employees ID? Explain briefly. 5
(CO2)
(PO3)
- c) Suppose R and S are two relations. In order to concatenate them (i.e. UNION operation) two conditions must be met. Briefly mention them. "Even these two conditions are satisfied the result of concatenation could be erroneous" - justify this fact with example. 10
(CO1)
(PO1)
2. a) Consider the following SQL with the assumption that mentioned table exists: 5
(CO1)
(PO1)

```
select id,name,dept
from students
where dept='CSE';
```

Your task is to write two equivalent queries of the above SQL statement using relational algebra.

- b) Consider the following entities (pk denotes primary key while $fk[x]$ implies foreign key referencing entity x): 5 × 3
(CO1)
(PO1)

```
Depts (NAME (pk), LOCATION, SIZE)
Students (ID (pk), NAME, CGPA, DEPT (fk [Depts]))
Employees (ID (pk), NAME, DEPT (fk [Depts]), JOININGDATE)
```

Note: Some employees may not be assigned to any department yet.

Write the standard SQL queries for the followings (query must be case in-sensitive where applicable):

- i. List of the department name and location whose name begins with letter 'C' and ends with 'E'.
 - ii. List of the employees name and designation along with their corresponding department size.
 - iii. List of all employees name and designation along with their corresponding department size if it is available otherwise it will show null (i.e. blank). *
 - iv. List of department and its location whose total number students is at least 400.
 - v. List of employees name and designation who are working for more than 5 years.
3. a) There is a danger of using *natural join* construct in SQL. Briefly explain it using example. 5
(CO1)
(PO1)
 - b) Differentiate between anonymous and named blocks in PL/SQL. "A function in PL/SQL can return more than one value." Justify it using suitable example. 5 + 5
(CO1)
(PO1)

- c) Explain the concept of total and partial participation in E-R diagram. Consider a one-to-many mapping cardinality between entities E_1 and E_2 . In implementation phase, is it possible to ensure total participation for both E_1 and E_2 using any table-level constraint? Justify your position. 10
(CO2)
(PO3)

4. a) Consider the following system description: 5 + 10
(CO3)
(PO2)

In IUT students are enrolled against a particular department. The Student Information System (SIS) maintains students basic information such as ID, Name and CGPA. Department name, its location are the relevant information for each department. Courses are offered in each semester. Each course has its code, title and credit hour. Multiple courses are taken by each student. Again one course can be taken by a group of students. The university runs a central board of different societies such as debate, photography and so on. For each society, SIS maintains the name and detailed description of the society. Each student must be attached to exactly one society.

Design the system using standard the E-R Diagram and write the DDL statements to implement it.

- b) Consider the following records of STUDENTS Table: 10
(CO1)
(PO1)

Table 1: STUDENTS (Data for Question No. 4 b)

ID	Name	CGPA	Dept
1	a	3.5	CSE
2	b	3.7	EEE
3	a	3.5	CSE

Your task is to split the table into two for both (i) lossless and (ii) lossy decomposition.

5. a) Define functional dependency. What is the basic motivation of using functional dependency in database design? Present example of each type of the following functional dependencies: 2 + 8
(CO1)
(PO1)
- i. trivial ii. non-trivial iii. semi-trivial iv. transitive
- b) Briefly mention primary rules of Armstrong's Axioms. Why are they called complete and sound? 5 + 5
(CO1)
(PO1)
- c) Is 2NF stricter than 3NF? Briefly explain. Consider the following records of CITIZENS table: 2 + 8
(CO1)
(PO1)
- (Note: the last 3 attributes are namely districts name, size and population)

Table 2: CITIZENS (Data for Question No. 5. c)

ID	Name	DistName	DistSize	DistPolulation(in million)
101	a	Dhaka	Large	20.5
102	b	Rajshahi	Medium	10.2
103	c	Dhaka	Large	20.5
104	a	Khulna	Medium	8.9
105	f	Rajshahi	Medium	10.2

Your task is to verify if the relation is in 3NF. You need to deduce functional dependencies from the given data set. If this is not in 3NF then suggest a loss-less decomposition. (Hint: You do not need to check 2NF condition, just check if it exhibits transitivity dependency for non prime attributes)

6. a) Briefly explain why the closure of attributes and concept of superkey are closely linked. Consider the following relation R and functional dependencies (FDs):
 $R(A, B, C, D)$ given FDs: $AD \twoheadrightarrow BC$ $B \twoheadrightarrow D$ $C \twoheadrightarrow A$
 Show all steps to determine all possible candidate keys for the above relation. 10
 (CO1)
 (PO1)
- b) Consider the following relation R and functional dependencies (FDs):
 $R(A, B, C, D, E)$ given FDs: $A \twoheadrightarrow C$ $C \twoheadrightarrow E$ $B \twoheadrightarrow D$ $E \twoheadrightarrow A$
 Show each step to verify if the given relation is in BCNF. 10
 (CO3)
 (PO2)
- c) Briefly present a comparative analysis of sparse and dense indexing. Consider the following citizen relation: 5 + 5
 (CO2)
 (PO2)
 $CITIZEN(ID, Name, BloodGroup, YearlyIncome)$
 Here $YearlyIncome$ indicates the exact yearly income of a citizen. Is it wise to apply bit-map indexing on $YearlyIncome$ directly? Justify your position.

Islamic University of Technology
 Organisation of Islamic Cooperation (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

WINTER SEMESTER, 2018-2019

Duration: 3 Hours

Full Marks: 150

CSE 4307: Database Management Systems

Programmable calculators are not allowed. Do not write anything on the question paper. There are 8(eight) questions. Question No. 7 and 8 are compulsory to answer. Answer any 4(four) from the remaining questions. Figures in the right margin indicate marks.

1. (a) One of the major drawbacks of traditional file processing system is the *difficulty to access data*. Explain it with a suitable example. Mention some widely used applications of modern database systems. [10]
- (b) A major purpose of a database system is to provide users with an abstract view of the data. Explain the concept of different types of *data abstraction* in modern database systems. [10]
- (c) What are projection and selection? Does the order of these operations matter in the final result? Explain with example. [05]
2. (a) Define and differentiate between Data-Definition Language (DDL) and Data-Manipulation Language (DML). [05]
- (b) Define super key, candidate key, primary key and foreign key. Consider the following system description of a typical library automation: [5+10]

There are books on different subjects such as Computer Science, Mechanical Engineering, Education etc. Each book have information such as Book Title, Author Name, Publisher Name, Publishing Year. There are a number of copies of a single book (e.g Database System Concepts, 100 copies). There are a number of departments in the University. Each department runs a number of programs. The students information system stores basic information of each students such as name, date of birth, address, father name, department name (i.e. CSE, EEE), program name (i.e. B.Sc. or M.Sc.). Students can borrow book and return book.

Your tasks are:

 - i. Design the E-R diagram and its equivalent DDL statements.
 - ii. In each case determine the primary key and foreign key.
- (c) What is the difference between inner join and outer join? Explain left outer join and right outer join with suitable examples. [05]
3. (a) Followings are the requirements of some tables. Each field is separated by comma (,) and additional requirements are stated in brackets []. [10]
 - **Departments**(ID primary key, Name of dept [Can not be empty], Establishment Year)
 - **Employees**(ID primary key [it is exactly a 9-digit number without any decimal part], Name [20 characters long but it can be in any language other than English], Date of birth [must not be empty and it can be less than 01-01-1960], Dept [foreign key of departments entity and it can not be empty], Blood Group [must be any one from A+, A-, B+, B-, AB+, AB-])
 - **Salary**(EmpID [foreign key of Employees entity], Date of Payment, Amount)

Your tasks are:

 - i. Create the tables using standard SQL.
 - ii. Now you forgot to include Salary (per month) and Join Date of the Employees. How can you solve this problem?

- (b) Write the SQL for the following queries: [10]
- Find out the name and date of birth of all Employees who joined in the last 3 months (i.e. 90 days).
 - Find out the name, date of birth and name of the department of all Employees who joined in the last 3 months (i.e. 90 days).
 - Generate a list containing department name and its total employees.
 - Generate a list containing department name and its total employees but include the department with at least 20 employees.
 - Generate a list containing department name and its total employees but include the department with at least 20 employees and whose departmental average salary is greater than 20000.
- (c) What is cardinality? Briefly explain different types of cardinality. [05]
4. (a) Null value introduces a number of problems in arithmetic operations in SQL statements. Justify with suitable example. [10]
- (b) What is jdbc? Consider the entity **Emp**(ID, Name, Address, Dept). Write a java program that will print the name, address and dept of those employees who work in 'Accounts' department. Assume that jdbc driver is already loaded. Write only the relevant part of the code. [10]
- (c) There are four integrity constraints on single relation. Name them with suitable examples. [05]
5. (a) What is the domain of attribute? Explain different types of attributes used in E-R model. [05]
- (b) Consider the following description: [10]
- XYZ is a large company comprising about 5000 employees. To automate the Human Resource 2 programmers are hired. Mr. Simple, one of the programmers, designs the ID of employee as simple numbers such as 000001, 000002 and so on. Mr. Complex, another programmer, designs the ID for the format: DEPTCODE-DESIGNATIONCODE-JOINDATE-SEQUENCENUMBER, (i.e. ACCOUNTS-MANAGER-01JAN2017-001).*
- Your tasks are:
- Criticize the above designs mentioning the strength and weakness of each option.
 - Propose your solution minimizing the weaknesses you have just mentioned.
- (c) Classify the constraints on generalization or specialization based on the followings: [10]
- Attribute of higher-level entity determines lower-level entity membership
 - The number of branching in its lower-level entity
 - Completeness
6. (a) What is the purpose of Normal Forms in database design? Explain the concept of 1st Normal Form with example. [05]
- (b) Mention the conditions of Boyce Code Normal Form (BCNF). Cite two examples both valid and invalid BCNF schema. Also mention how you will decompose an invalid BCNF into several valid BCNF schema. [10]
- (c) Consider the following schema and records (as presented in table 1) for a student information management system: [10]
- students*(ID, Name, Date of Birth, Country Name, Capital Name, Total Population)
- Your tasks are:
- Verify if the given schema is in BCNF.
 - If it is not in BCNF then decompose it as directed by the BCNF algorithm.

Table 1: Records for Question No. 6.(c)

ID	Name	DOB	Country_Name	Capital	Population (in m)
1	a	1-1-87	Bangladesh	Dhaka	160
2	b	1-2-85	Afghanistan	Kabul	32
3	c	1-3-81	Bangladesh	Dhaka	160
4	d	1-3-81	Cameroon	Yaounde	17
5	e	1-4-77	Afghanistan	Kabul	32

7. (Compulsory)

Consider the following scenario:

ABC is a large bank with few hundreds of branches located at different parts of the country. Customers must provide their profile information such as Name, Date of Birth, Address before opening any account. Once a customer's profile is available he/she can open multiple accounts reusing the profile information. After opening account regular transactions are made. There are two types of transactions such as withdraw and deposit.

Bank offers loans. Only existing customers who have valid accounts are primarily eligible for getting a loan. There are 3 types of loan schemes such as Platinum, Gold and Silver. Each loan has its own properties as described in table 2.

Table 2: Properties of loan schemes for Question No. 7

Loan Scheme	No. of Instalment	Interest Rate (per year)	Eligibility
Platinum	100	5	Total Transaction (i.e. add both types of transactions) in the last 12 months must be ≥ 2000000
Gold	75	8	Total Transaction in the last 12 months must be between 2000000 and 1500000
Silver	50	12	Total Transaction in the last 12 months must be between 1500000 and 1000000

Your tasks are:

(a) Design the table definitions and issue the required DDLs. Additional assumptions are welcome in design phase. [08]

(b) Write a function to assign a customer to a specific category of loans as mentioned. (assume that each customer makes regular transactions such as deposit and withdraw). [08]

Input: Account No. Output: Rejected or Accepted, if Accepted it should also show which type of loan can be granted based on the Eligibility parameter as described in table.

(c) Once a customer is assigned to a specific loan scheme, write a procedure to schedule the loan. Assume each loan must be paid after 6 months interval. [09]

Input: Account No, Loan Scheme, Total Amount, Starting Date. Output: It will schedule x number of equal instalments based on Total Amount and No. of Instalment of that particular loan scheme. The schedule information should include: account no, loan scheme, Instalment no.(starts from 1 to x), Instalment amount (i.e. total amount will be equally distributed), Payment Date (i.e. after every 6 months from the Starting Date), Payment Status. All fields except Payment Status should be initialised by the procedure. Payment Status should be set to null.

8. (Compulsory)

- (a) Suppose you have issued an UPDATE statement to a table *citizens* as mentioned in Question No. 8.(c) with some WHERE CLAUSE (as you like). [05]

Write an anonymous block that will execute the above UPDATE statement. If none of the records are affected by the statement then it will print *NO RECORDS ARE UPDATED*, otherwise it will count the total number of updated records (i.e. X) and will print *A TOTAL OF X RECORDS ARE UPDATED*.

- (b) What is the basic difference between row-level trigger and statement-level trigger? Explain with example. [05]

- (c) Consider the table *citizens(id, name, dob, salary)*. The Government of Bangladesh (GOB) has created one fund of total BDT *totalAid_amount*. [15]

GOB wants to ensure (but can not guarantee) each citizen receives an amount *gob_allowance* such that after receiving it his/her total earning (i.e. *salary + gob_allowance*) is equal to the average income of the country (average is computed before any *gob_allowance* is given). The citizens having more than the average salary of the country are not eligible for this scheme.

For this purpose GOB invites applications from needy and interested people. The applications are stored in *applied(citizen_id, date of application)* table (assume only the valid persons apply). The citizens who have not applied will not be considered even his/her salary is very low.

The citizen (who applied) with the lowest salary will get the highest priority to receive *gob_allowance* and *gob_allowance* amount is determined by the difference of his/her salary and average salary of the citizen. GOB can not ensure sufficient fund for all needy citizens. So the process terminates whenever the fund is exhausted (i.e. *totalAid_amount=0* or *totalAid_amount* is less than the difference of the average salary and the salary of the particular citizen).

When a citizen receives *gob_allowance* an appropriate update of *citizens* table should be made.

- Your task is to write a procedure *distribute_allowance* satisfying the above requirements. The procedure will take only one IN parameter i.e. *totalAid_amount*. [Hint: use explicit cursor to select the candidates as per the description]

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4307: Database Management Systems

Programmable calculators are not allowed. Do not write anything on the question paper.
 Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. a) Briefly explain the "Integrity Problem" in a traditional file processing system. 5
(CO1)
(PO1)
- b) What is domain of attribute in relational database? How are they ensured in DDL statements? Explain with example. 5
(CO1)
(PO1)
- c) Define Super Key, Candidate Key, and Primary Key. Use data from Table 1 to deduce super key, candidate key, and primary key. Show each step. 10
(CO1)
(PO1)

Table 1: Data for Question 1.c)

Name	Address	Dept	CGPA
a	x	CSE	3.50
b	y	CSE	3.60
a	z	EEE	3.50
c	x	MPE	3.40

2. Followings are the requirements of 2 relational tables:

• Table Name: **Divisions**

Attribute	Meaning and Requirement
DID	Primary key, exactly 6 digits with no decimal part.
DName	50 Characters irrespective of language setting, can not be empty.
DSize	Value in Square KM, can not be less than 100.

• Table Name: **Persons**

Attribute	Meaning and Requirement
PID	Primary key, it is exactly a 6-digit number without any decimal part.
Name	20 characters long irrespective of language.
DOB	Date of Birth, must not be empty.
Division	Foreign Key referencing Divisions and it can not be empty.
Blood Group	Must be any one from A+, A-, B+, B-, AB+, AB-.
Salary	Monthly salary, must be greater than 5000.

- a) Create the tables using standard SQL. 10
(CO1)
(PO1)

b) Write the SQL for the following queries:

- List the division name and size in ascending order of their size.
 - List the person ID and name who live in divisions that start with 'D' and ends with 'A'.
 - List the person ID and name who live in division with the largest size.
 - List the division name and its total inhabitants.
 - List the division name and its total inhabitants where total inhabitants is at least 50000.
- c) The main motivation for the concept of foreign key is to remove "redundancy" and "inconsistency" in relational database. Explain it using suitable example.

5
(CO1)
(PO1)

10
(CO1)
(PO1)

3. Consider the following scenario:

Citizen Information: It stores information for all citizens of the country including ID, Name, Date of Birth (DOB), Blood Group (BG), and Profession. Professions are organized as 2-layer hierarchy to include both generic and specific name of the profession. For instance, a "Civil Engineer" is a specific name for the category of the profession "Engineer".

Driving License and Accidents: Each citizen may have at most one driving license having the basic information such as Name, DOB, Type of Vehicle (i.e., Light or Heavy), Issue Date, Expiry Date. (Note that a certain level of redundancy is acceptable here.) Whenever any accident occurs, it is recorded with short description, date, and time, ID of the driver and the place of the accident.

a) Draw the Entity Relationship Diagram (ER-D) using standard symbols.

15
(CO2)
(PO3)

b) Implement the ER-D using standard DDL statements.

15
(CO1)
(PO1)

4. a) Considering your design in Question 3, write a PL/SQL function using the following information:

10
(CO1)
(PO1)

Input	Driver ID (or similar)
Output	Status (i.e., Good, Average, Bad)
Algorithm	If the total number of accidents by this driver during the last 6 months exceeds 10, then his/her status is "Bad". If it is between 2 to 10 then it is "Average". Otherwise it is "Good".

b) Present a suitable example to differentiate between row-level and statement-level trigger.

5
(CO1)
(PO1)

5. a) "Trivial and Semi-Trivial functional dependencies are always true." Justify your position regarding the statement using suitable example data.

5
(CO1)
(PO1)

b) Mention the primary rules of Armstrong's Axioms. Why are they called complete and sound?

5
(CO1)
(PO1)

c) Consider the relation $R(A, B, C, D)$ and the following functional dependencies (FDs):

$BC \rightarrow AD$ $D \rightarrow B$ $A \rightarrow C$

Show the steps to determine all possible candidate keys for the above relation. Also, show if the given relation is in BCNF.

15
(CO1)
(PO1)

- a) Define prime attribute. Consider the relation $R(A, B, C, D)$ and the following functional dependencies (FDs):
 $AC \rightarrow BDEF$ $CD \rightarrow E$
 Show the steps to determine if the given relation is in 3NF. 10
(CO1)
(PO1)
- b) What is the basic motivation of multi-level indexing in database? Consider the sequences/-timings (i.e. $t_i < t_{i+1}$) of data insertion in a B Tree shown in Table 2. Use the concept of m-way binary tree where $m = 4$. 10
(CO1)
(PO1)

Table 2: Data for Question 6.b)

Timing	Data points (Key values) for insertion
t_1	9,18,42,51
t_2	65,70,81
t_3	28,36
t_4	7,14

Construct the B Tree showing each step. Also, highlight the modification needed to convert it into a B+ Tree.

- c) Briefly explain the major characteristics of Big Data. What is a "blind zone" in this context? 10
(CO1)
(PO1)